

Dataset & Guideline-Source Setup Guide

RAG-Based Clinical Decision Support / Triage Assistant

This covers exactly what to source, how to get it, how to prep it, and what to evaluate against — so week 1 isn't spent guessing at data.

1. Grounding Corpus (what the assistant retrieves from)

This is the most important choice you'll make — the whole point of RAG is that answers are only as good as what's retrievable. Don't build your own guideline scraper in week 1; strong RAG-ready corpora already exist.

Primary recommendation: EPFL-LLM Guidelines Corpus

`<cite index="6-1">`This is a curated collection of clinical practice guidelines (CPGs) comprising 48,096 articles from 17 globally recognized sources, spanning internal medicine, pediatrics, oncology, infectious disease, and more, across high- and low-resource settings.`</cite>` It was built specifically for medical LLM training/RAG use, is already chunked and structured, and is free on Hugging Face ([epfl-llm/guidelines](https://huggingface.co/epfl-llm/guidelines)).

Why start here: it's already vetted, multi-domain, and RAG-ready — you skip weeks of scraping/cleaning and go straight to building your retrieval pipeline.

Supplementary corpora (add if you need broader coverage)

- `<cite index="4-1">`MedWiki — a curated, medical-only subset of Wikipedia (Dec 2022 snapshot) containing about 1.3M chunks from more than 150K articles, roughly 3% of the original encyclopedia, released specifically as a RAG-ready dataset.`</cite>` Good for general medical background knowledge outside strict clinical guidelines. ([hf.co/datasets/jpcorb20/medical_wikipedia](https://huggingface.co/datasets/jpcorb20/medical_wikipedia))
- `<cite index="4-1">`ClinicalCorp — a larger assembled corpus for clinical-domain RAG containing over 2.3M chunks, combining guideline sources with other clinical text.`</cite>` Use if your triage assistant needs to reason over broader clinical literature, not just formal guidelines.
- `<cite index="4-1">`A separate RAG-ready release of the open-source guideline set used to pretrain the Meditron medical LLM, containing more than 710K chunks across eight open-source guideline datasets`</cite>` — a good cross-check/second source to validate retrieval quality against.

If you want a smaller, more curated set instead

Direct sources you can pull specific guidelines from (better if your triage assistant is scoped to a narrower specialty, e.g., primary-care triage only):

- WHO guidelines** ([who.int/publications](https://www.who.int/publications)) — global, freely downloadable PDFs
- NICE guidelines** ([nice.org.uk](https://www.nice.org.uk)) — UK, freely accessible, well-structured
- CDC clinical guidance** ([cdc.gov](https://www.cdc.gov)) — US, freely accessible

Recommendation for your timeline: start with `epfl-llm/guidelines` as your full corpus (week 1-2), and only narrow to a specialty-specific subset later if evaluation shows retrieval precision suffers on a broad corpus. Don't pre-narrow before you've measured the problem.

2. Evaluation Datasets (what you test hallucination-rate and accuracy against)

Your paper's credibility depends on rigorous evaluation, not the pipeline — so this list matters as much as the corpus.

Dataset	What it tests	Use for
MedQA (USMLE-style questions)	Clinical reasoning, multiple-choice	Core accuracy benchmark — most-cited comparison point in the literature
PubMedQA	Yes/no/maybe biomedical research QA	Tests grounding against research literature specifically
MedMCQA	Large-scale medical entrance-exam MCQs	Broader coverage, good for statistical significance with more test items
MEDIQA-CORR	Medical error detection & correction in clinical notes	<cite index="4-1">Directly tests whether your system catches and corrects medical errors — a strong hallucination/safety-relevant benchmark, used in a 2024 shared task with an open leaderboard you can compare against.</cite>
MultiCaRe	<cite index="9-1">Real de-identified clinical case reports (96,428 cases from 75,382 open-access PubMed Central articles, 1990–2023), including images and captions.</cite>	Good for realistic, messy triage-style case scenarios rather than clean exam questions — use for qualitative case studies in your paper

Recommended sequence: validate your pipeline on MedQA first (widely used, easy to compare against published baselines), then run PubMedQA and MEDIQA-CORR for your hallucination/safety-specific claims, and use a handful of MultiCaRe cases as illustrative qualitative examples in your paper/demo.

3. What "Hallucination Rate" Actually Means Here — Eval Protocol

Don't just report accuracy — the 2025 JAMIA review of RAG-in-clinical-LLM literature is explicit that this is where most systems fall short: <cite index="1-1">Studies increasingly fine-tune LLMs specifically to assess the relevance of retrieved evidence, verify that

generated statements are evidence-based, and confirm the response actually addresses the clinical question — and despite RAG's benefits, there remains real room for improvement in how rigorously this is evaluated.</cite>

Concretely, measure and report all of:

1. **Answer accuracy** (vs. MedQA/PubMedQA ground truth)
2. **Groundedness** — does every claim in the generated answer trace back to a retrieved passage? (manually annotate a sample, or use an LLM-as-judge with spot-checking)
3. **Retrieval precision** — are the top-k retrieved chunks actually relevant to the query?
4. **Abstention behavior** — does the system correctly say "I don't have enough grounded information" rather than guessing, when the corpus doesn't cover the question? (This is an underrated, easy differentiator most student RAG projects skip.)

This four-part protocol is a stronger, more publishable evaluation section than accuracy alone — build it into your week 5-6 plan, not as an afterthought in week 15.

4. Preprocessing & Chunking Recommendations

- **Chunk size:** start at 256–512 tokens per chunk with ~10–15% overlap — standard starting point for clinical guideline text, which tends to have dense, self-contained recommendation statements.
 - **Chunk by structure, not just token count where possible:** guideline documents usually have clear section headers (e.g., "Diagnosis," "First-line treatment," "Contraindications") — preserve these as metadata on each chunk so you can filter/boost retrieval by section type later.
 - **Keep source citations attached to every chunk** (source guideline name, publication year, section) — this is what lets your system show "grounded in [Guideline X, 2023]" in its answers, which is both a safety feature and a nice demo moment for reviewers.
 - **De-duplicate across corpora** if you combine EPFL-guidelines + MedWiki + ClinicalCorp — overlapping content across sources will otherwise skew retrieval toward duplicated topics.
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5. Vector Database & Retrieval Stack

Given your 3–4 month timeline and existing RAG experience, don't over-engineer this layer — pick one and move on:

Component	Recommendation	Why
Embedding model	A medical-domain-tuned sentence embedding model (e.g., a BioBERT/PubMedBERT-based sentence encoder) over a generic one	Domain-tuned embeddings measurably improve retrieval precision on clinical text vs. general-purpose embeddings
Vector DB	Chroma or Qdrant (both free, local-first, fast to set up)	You don't need a managed/cloud vector DB for this scale of corpus (hundreds of thousands of chunks is well within local-first tooling)
Retrieval strategy	Hybrid (dense + keyword/BM25) rather than dense-only	Clinical text has exact-match terms (drug names, dosages) where keyword search outperforms pure semantic search — hybrid retrieval is a legitimate, citable design choice for your Methods section
LLM (generation layer)	Whatever API you already have experience with (per your prior projects)	Don't burn week-1 time re-learning a new provider; consistency with what you know speeds the whole build

6. Data Governance Note (say this explicitly in your proposal)

Every source above is either open-access/de-identified (MultiCaRe, PubMed Central) or a formal public guideline document (WHO/NICE/CDC, EPFL corpus) — **no real patient-identifiable data is used anywhere in this project**. State this plainly in your proposal; it preempts the most common review-panel question for any "clinical" capstone and shows you've thought about it rather than being asked and improvising an answer.

7. Week 1–2 Action Checklist

1. Download ([epfl-llm/guidelines](#)) from Hugging Face; inspect structure and a sample of ~20 chunks manually to understand formatting.
2. Set up Chroma or Qdrant locally; embed and index a small subset (e.g., 5,000 chunks) as a pipeline smoke test before indexing the full corpus.
3. Download MedQA and PubMedQA; hold these out entirely from anything used in corpus-building — they're evaluation-only.
4. Implement the four-part eval protocol (accuracy, groundedness, retrieval precision, abstention) as a script now, even before the generation pipeline is fully built — you'll reuse it every week after.
5. Draft the data-governance paragraph for your proposal (Section 6 above) — this is a five-minute task now that saves a review-day scramble later.

Ready for the week-by-week build plan next (mirroring the FL doc's format), or do you want to start on the formal proposal doc now that the data foundation is settled?